

Full Length Article

Scanning strategies effect on temperature, residual stress and deformation by multi-laser beam powder bed fusion manufacturing

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ARTICLE INFO

Keywords:

Powder bed fusion

Scanning strategies

Residual stress

Deflection

Dual laser beams manufacturing

ABSTRACT

Laser beam Powder Bed Fusion (PBF-LB) is an increasingly attractive technology due to its capability of manufacturing complex and customised metal parts. However, the build cycle time and resulting residual stress profiles are limiting a wider uptake of PBF-LB manufacturing. The incorporation of additional laser beams in a single build chamber, i.e. multi-laser beam Powder Bed Fusion (PBF-MLB) technology, has been proposed as the next generation of PBF-LB technology. A series of computational process simulations of PBF-MLB was carried out using a 3D coupled thermomechanical model to investigate optimum multi-laser scanning strategies. The influences of twelve different scanning strategies on temperature, the final residual stresses, the z- (build) direction deflections by dual Gaussian profile laser beams PBF-MLB were investigated. It was found that the 90° layer rotation is necessary to reduce the residual stress, and a 'dual-laser following but time-delayed' scanning strategy had the lowest final residual stress. Another scanning strategy (45° rotation approaching beam scanning) resulted in the lowest z-direction deflection for the dual laser beams PBF-MLB manufacturing. These findings provide insight and application of PBF-MLB finite element process simulation in improving industrial additive manufacturing of metal components.

1. Introduction

Laser beam Powder Bed Fusion (PBF-LB) Additive Manufacturing (AM) produces components by consecutively melting thin layers of powder to form 3D objects. Within each layer, a finely rastered laser beam creates a micro-size melt pool as it traces the 2D shape in the powder bed. In this layer-by-layer process, the melt pool rapidly solidifies as the laser spot moves on. The repeated melting and solidification process causes a complex thermal history and thus determines the resulting residual stress and deformation of the manufactured components. Multiple industries, such as medical device and aerospace, are rapidly adopting PBF-LB as a manufacturing method and thus are faced with addressing this stress and deformation challenge [1,2].

Another major issue limiting the wider adoption of PBF-LB is the productivity rate and associated costs [3–5], particularly for larger components. Strategies to improve the PBF-LB build rate include modifying process parameters, such as enlarging the laser beam diameter, layer thickness and laser power to be 1 mm, 200 μm and 1000 W, respectively. However, such methods can result in 10 % loss of density

and poor component geometry accuracy [6,7]. An alternative strategy to add additional laser beams (PBF-MLB) has the potential to combine high manufacturing efficiency and high precision [7].

The dual laser beams PBF-MLB manufacturing is not only effective in improving the build rate and reduced cycle times [7,8] for standard PBF parts, but also makes it more practical for larger macro scale components. In dual laser beams PBF-MLB manufacturing, the total exposure time can be shortened up to a factor of 2 compared with the single laser beam PBF-LB manufacturing technology while the production rate could be improved to ~ 60 cm³ per hour, comparable to that of the electron beam powder bed fusion manufacturing (PBF-EB) [4].

Most computational modelling studies PBF-LB found in the literature, focus on single laser beam processes. Computational process modelling can provide insight into the optimum PBF-MLB configuration and process parameters, enabling consistent high build quality, greater part complexity and design freedom [9]. The ability to accurately simulate the physical PBF-MLB process and predict the thermal history and the final residual state stress would constitute a considerable

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Nomenclature			
ρ	Density of solid, kg/m ³	v	Laser scanning speed, m/s
C_p	Specific heat, J/(kgK)	q	Heat flux, w/m ²
k	Thermal conductivity of solid, w/(m ² K)	q_{conv}	Convection heat flux, w/m ²
∇T	Temperature gradient, K	h	Heat transfer coefficient, w/(m ² K)
Q	Energy density, w/m ³	T_s	Surface temperature, K
L	Latent heat, J/kg	T_r	Chamber temperature, K
A	Heat source absorption coefficient	q_{rad}	Radiation heat flux, w/m ²
P	Laser beam power, w	ε	Emissivity coefficient
R	Heat source spot radius, m	σ	Stefan-Boltzmann constant, w/(m ² K ⁴)
r	Distance to the laser beam centre, m	α	Thermal expansion coefficient, 1/K
H	Hatch spacing, m	L_h	Layer height, m

advancement in AM and further empower product and process design.

Two kinds of the PBF-MLB technologies were developed [10]: single trajectory lasers with multi-beam and multi trajectory lasers with multi-beam. Single trajectory lasers with multi-beam utilise the secondary beam as an auxiliary heat source. Heeling et al. [11] noted that the secondary beam could heat up the region surrounding the melt pool, thus decreasing the cooling speed. Multi trajectory lasers with multi-beam involve all beams acting as powder melting heat sources. Li et al. [7] studied the resulting microstructure and mechanical properties when the laser overlap was varied during the dual laser beams PBF-MLB and found that PBF-MLB could obtain uniform part properties. Heeling et al. [3] investigated the influences of scanning strategies on the melt pool dimensions, dynamics and temperature gradients in the PBF-MLB and the results showed that PBF-MLB could reduce the temperature gradients by up to 20 %. Masoomi et al. [8] reported a dependence of temperature, microstructure, defect occurrence and mechanical properties on scanning strategies in PBF-MLB, which indicated that temperature gradients were highly sensitive to scanning strategies and the number of laser beams. Commercial development of multi-beam systems is currently underway, including PBF-MLB technologies from SLM Solutions [12], Renishaw [13], EOS [14] and Concept Laser [15].

It is known that scanning strategies influence the temperature distribution, residual stress and distortion in the single laser beam PBF-LB manufacturing [16,17]. The dynamic microscale laser spot and layer-by-layer feature of PBF-LB manufacturing results in a non-uniform

temperature distribution, residual stress and deflection [18,19]. The study by Li et al. [17] revealed that the horizontal and vertical sequential scanning strategies generated a relatively uniform residual stress distribution in the x and y direction. Robinson et al. [19] experimentally demonstrated that the XY alternative scanning strategy (where the scan direction rotates 90° after finishing one layer) caused a more uniform and lower residual stress compared with no layer rotation. Masoomi et al. illustrated island scanning is an effective strategy to reduce the final component residual stress [20].

The part deformation and cracking influenced by the high residual stress are among the main defects of additively manufactured parts, which can significantly influence the in-service performance of the printed parts [17]. In addition, deflection influences the final parts quality and performance [21], which is one of the factors that affect the application of PBF-LB [3]. Cheng et al. [16] numerically studied the influences of eight scanning strategies on temperature, stress and deformation by the single laser beam PBF-LB and the results showed that the 45° line scanning had the smallest build direction deformation compared with other scanning strategies and the island scanning produced the maximum peak temperature due to the residual heat effect within the short scanning path.

While there has been considerable progress in optimising single laser beam PBF-LB scanning strategies, the optimum PBF-MLB scanning strategy and laser interaction remain unclear [3,8]. With the proven link between scanning strategy – residual stress – and part deflection

Table 1

Ti6Al4V Material properties [22–24].

Temperature (K)	Density (kg/m ³)	Specific heat (J/kg/K)	Thermal expansion coefficient $\times 10^{-5}$ (1/K)	Thermal conductivity (W/m/K)	Elastic Modules $\times 10^4$ (MPa)	Yield strength (MPa)
293	4421	580		10.07		1040
303					11.16	
333			0.87			
373						940
423					11.16	
473		680		12.02		852
603						762
623					11.11	
673		760		15.03		
713						661
823					10.49	538
983			1.06			
1023					9.20	
1073	4300.5	800		17.05		
1088						64
1123						53
1223					7.50	21
1363						6.81
1423					5.68	
1623					3.88	
1823					2.15	
1893	4202	930		19.97		
1928	3801	950	1.35	60		

[20], it is hypothesised that the addition of a second laser (and thus more energy) could be designed to further reduce residual stress and deflection, thus making PBF-MLB more suitable for manufacturing than PBF-LB.

The objective of the current work is to computationally simulate the thermo-mechanical process of the dual laser beams PBF-MLB and investigate the influences of scanning strategies on the process temperature, the resulting residual stress and deflection.

2. Methods

2.1. Thermal mechanisms

The temperature dependent Ti6Al4V material properties were chosen due to the expected large temperature gradients in the model. The thermal and mechanical material properties (conductivity, density, specific heat, elastic modulus and yield stress) for both the solid part and the base plate were obtained from the literature [22,23] and are shown on Table 1. The Ti6Al4V powder properties were assumed to be dependent on solid material properties via powder bulk porosity [22] as shown in our previous publications [23,24]. All material properties were considered to be isotropic and homogenous. The governing equation for the heat transfer process is formulated by using the energy equation [23,25]:

$$\rho C_p \frac{dT}{dt} + \rho \frac{d(fL)}{dt} = -\nabla \cdot q + Q + \dot{D}_{mech} \quad (1)$$

where ρ is the density, C_p is the specific heat, T is temperature, t is time, L is the latent heat of fusion, q is the heat flux, Q is the energy density of laser heating and $\dot{D}_{mech}$ is the thermo-mechanical dissipation caused by plastic deformation [23,26,27]. Here, the latent heat which describes the solid to liquid phase transition of the material was set as 370,000 J/kg [22]. The liquid fraction is calculated by [28]:

$$f = \begin{cases} 0 & T < T_s \\ \frac{T - T_s}{T_L - T_s} & T_s \leq T \leq T_L \\ 1 & T > T_L \end{cases} \quad (2)$$

where T_s and T_L are the solidus and liquidus temperatures, respectively.

In this study, the PBF-MLB thermal cycles involve the laser beams heating process and the cooling process. For the laser beams heating process, the Gaussian heat profile (Fig. 1) is a widely adopted model to mathematically formulate the profile of laser heat source [29–31] and it is employed in this work. The energy density is assumed to follow the Gaussian distribution [29]:

$$Q = \frac{4\sqrt{2}AP}{\pi\sqrt{\pi}R^3} e^{-\frac{2r^2}{R^2}} \quad (3)$$

where A is heat source absorption coefficient, P is laser beam power, R is laser spot radius and r is the distance to the laser beam centre. The

thermo-mechanical dissipation $\dot{D}_{mech}$ is defined as [27]:

$$\dot{D}_{mech} = \sigma : \dot{\epsilon}^{vp} \quad (4)$$

where σ is the stress and $\dot{\epsilon}^{vp}$ is the visco-plastic strain. The thermo-mechanical dissipation $\dot{D}_{mech}$ accounts for a minimal contribution to heat generation in PBF-LB process [32].

The cooling process of the additively manufactured parts includes heat conduction to the surrounding solid material and base plate, part-powder interface conduction (approximated as free surface convection [23,33], active layer (newly added and undergoing heat flux) convection and radiation from the active layer top surface to the surrounding atmosphere until the finish of the final part. The thermal transfer mechanisms details have been shown on [23]. The influence of melt pool convection, radiation and evaporation were ignored due to the small size of the melt pool [17]. The heat flux due to conduction can be formulated as:

$$q = -k\nabla T \quad (5)$$

where k is the temperature-dependant thermal conductivity of solid material. The heat transfer boundary conditions of convection and radiation occur at the top surface of the active layer until the next layer was added [23]. The active layer convection (which is between the active layer and the surrounding atmosphere) heat dispersion can be expressed by Newton's law of cooling:

$$q_{conv} = h(T_s - T_r) \quad (6)$$

where h is the heat transfer coefficient of convection [34], T_s is the surface temperature of the workpiece and T_r is the chamber temperature [22]. Here, the characteristic heat transfer coefficient h was set as 12.7 W/(m²K) [35]. The active layer radiation heat loss (radiation from the active layer to the surrounding atmosphere) can be defined by Stefan-Boltzmann's law:

$$q_{rad} = \epsilon \sigma (T_s^4 - T_r^4) \quad (7)$$

where ϵ is the emissivity coefficient, σ is the Stefan-Boltzmann constant. Here the emissivity of the active layer surface ϵ and the Stephan-Boltzmann's constant σ were set as 0.35 [35] and 5.669×10^{-8} W/(m² K⁴), respectively [36].

2.2. Mechanical mechanisms

The fully coupled numerical thermal-stress simulations were performed using the general purpose finite element software ABAQUS FEA (Dassault Systems, USA, 2017). The equilibrium for the mechanical analysis is given by:

$$\nabla \cdot \sigma = 0 \quad (8)$$

where σ is the stress tensor. The mechanical constitutive law for the elastic problem is defined as:

$$\sigma = C : \epsilon_e \quad (9)$$

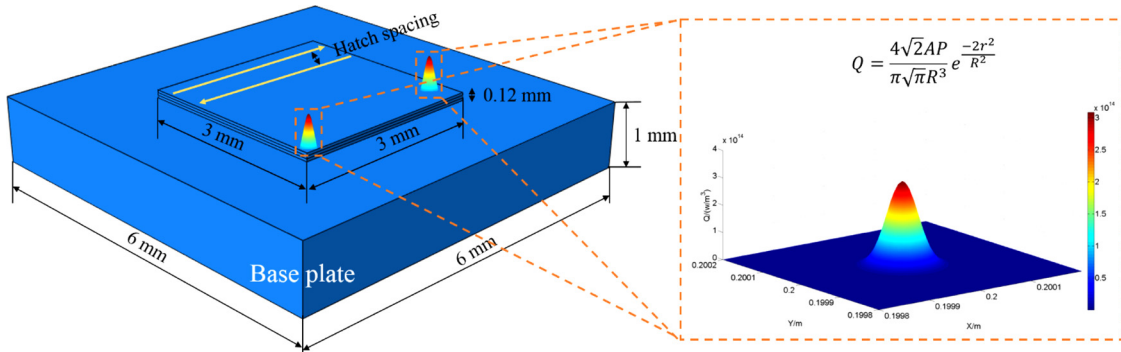


Fig. 1. Model configurations.

where $\mathbf{C}$ is the material stiffness tensor and $\mathbf{e}_e$ is the elastic strain tensor. The total strain can be represented as:

$$\boldsymbol{\varepsilon}_{Total} = \boldsymbol{\varepsilon}_e + \boldsymbol{\varepsilon}_p + \boldsymbol{\varepsilon}_T \quad (10)$$

where $\boldsymbol{\varepsilon}_{Total}$, $\boldsymbol{\varepsilon}_p$ and $\boldsymbol{\varepsilon}_T$ are the total, the plastic and the thermal strain, respectively. The thermal strain component is given by:

$$\boldsymbol{\varepsilon}_T = \alpha \nabla T \quad (11)$$

where α is the coefficient of thermal expansion.

2.3. Configuration of the case study

In this work, a rectangular additively manufactured part (3 mm × 3 mm × 0.12 mm) and a rectangular base plate (6 mm × 6 mm × 1 mm) were modelled (Fig. 1). The powder layer thickness was 40 μ m. The converged mesh size of 20 μ m × 20 μ m × 20 μ m (i.e. one - fifth of the laser beam diameter) was applied on the part while a relatively coarse mesh (i.e. 0.2 mm × 0.2 mm × 0.2 mm) was adopted for the base plate. Mesh-independent results were achieved by sampling the stress at the left bottom node of the solid part, and was found to be comparable to other similar scale models [20,37]. The total element number was 139,500. To obtain the temperature, stress and displacement information, the 8-node trilinear displacement and temperature mesh element (C3D8T) was chosen. The bottom of the base plate was fully constrained as the mechanical boundary condition during the whole process except for the base plate constraints releasing steps. All the nodes at the bottom of the base plate, except for the four extremity nodes, were unconstrained to allow the deformation of the part at the final steps [38]. In each step, the simulation models employ the *MODEL CHANGE* function in ABAQUS to simulate the addition of new layers and then Gaussian energy densities were applied to scan the designed paths of each layer.

The initial temperature of the whole system was set to be 293 K. The constant laser beam scanning speed, the heat source spot diameter and the hatch spacing were set as 0.6 m/s, 0.1 mm and 0.1 mm, respectively. The three layers were deposited sequentially with an interval of recoating time of 10 s, which is the time for a new layer to be spread out prior to laser scanning [16]. The complete process modelling parameters are summarised on Table 2. Based on the predicted melt pool depth of 60 μ m, trial builds and follow on μ CT imaging at 1.5 μ m resolution, the proposed process parameters will result in approximately fully dense parts.

2.4. Scanning strategies

To investigate the scanning strategies influence on the local thermal distribution resulted residual stress and deformation [16], twelve different scanning strategies were employed in the computational modelling (Fig. 2). Case (a) is the opposite *S* scanning, where the two laser beams move parallel to the bottom edge of the part (0°), parallel to each other but in opposite directions with a constant offset of hatch spacing *H* at any one time. The '*S*' scanning means the laser beam heat source reverses scan direction after finishing each hatch. Case (b) and case (c) are parallel *S* scanning, where the two laser beams move parallel to the bottom edge but in the same direction. For case (b), the laser beams scan direction rotates 90° for the next layer while no layer rotation for case (c). Case (d) is the 0° approaching beam scanning and the two laser beams scan from respective top and bottom sides and close to each other gradually while case (e) is the 45° rotation approaching beam scanning that the laser beams move with a 45° angle in terms of the *x* direction. Case (f) is the opposite halves scanning that two laser beams responsible for left and right halves of the model, respectively. Case (g) is the parallel halves scanning, where the two beams keep working parallel with a specific offset distance of half of the part width. Case (h) is the island approaching beam scanning, where two laser beams scan individual islands. Case (i) is the island mixed scanning that two laser

beams move parallel to each other but in opposite directions within each individual island. For island scanning of case (h) and (i), the whole scanning domain has been divided into 9 islands and rotates 90° for the sequence island scanning. Cases (j) and (k) are the contour fill scanning that case (j) scans the outside contour first while case (k) scans the part centre area first. Case (l) is the delayed scanning that the secondary laser follows the first laser and post-heating the powder bed, which aims to slow down the cooling rate along the laser moving direction. The delayed laser beam applied half of the first melting laser power and the constant offset was set as 90 μ m based on literature [11]. Except case (c), all the scanning strategies rotate 90° for the next layer.

An user defined subroutine DFLUX was programmed using FORTRAN to implement the moving dual laser beams by applying two Gaussian heat profiles of the laser beams to the powder bed according to the respective scanning speed and scanning strategies as shown in Fig. 2.

In order to establish overall trends due to increased number of lasers, three additional PBF process models were simulated for the above geometry using one, four and thirty lasers. In the four and thirty laser models, all laser beams were set to simultaneously scan the above part using the same opposite *S* scanning strategy (Fig. 2a) only. Thus two comparative studies were produced, first the optimum number of lasers for a given laser scanning strategy and second, the optimum laser scanning strategy for dual laser beams PBF-MLB.

3. Results and discussions

3.1. Influence of the number of laser beams on residual stress

In these computations, single laser beam, dual laser beams, four laser beams and thirty laser beams were separately employed as the heat source. Fig. 3 shows the computational modelling results of the Von Mises stress distributions for different numbers of laser beams by using the opposite *S* scanning pattern (Fig. 2a) at the interface of the part with the base plate. The study shows that maximum Von Mises stress for single laser beam, dual laser beams, four laser beams and thirty laser beams were 1438 MPa, 1379 MPa, 1303 MPa and 1201 MPa, respectively. The average Von Mises stress (the average value of all the nodes at the bottom surface) of the dual laser, four laser and thirty laser beams PBF-MLB were 3.39 %, 4.63 % and 9.53 % lower than that of the single laser beam PBF-LB. The maximum Von Mises stress and average Von Mises stress results (Fig. 3) indicate that to manufacture a single part, an increase in the number of laser beams reduces the magnitude and lower average residual stress distribution for the larger number of laser beams than that of the single beam PBF-LB modelling. The maximum *z* direction deflections for the single laser beam, dual laser beams, four laser beams and thirty laser beams PBF-MLB modelling were 3.61 μ m, 3.49 μ m, 3.26 μ m and 2.25 μ m, respectively. The lower residual stress and *z* direction deflection were formed

Table 2
Process parameters applied.

Symbol	Material	Ti6Al4V
	Base plate size (m)	0.006 × 0.006 × 0.001
	Base plate preheat temperature (K)	293
	Powder preheat temperature (K)	293
	Part size (m)	0.003 × 0.003 × 0.00012
L_h	Layer height (mm)	0.04
	Cooling step time each layer (s)	10 [16]
p	Laser beam power (w)	95 [39]
A	Heat source absorption coefficient	0.4
R	Heat source spot radius (mm)	0.05 [40]
v	Laser scanning speed (m/s)	0.6 [41]
H	Hatch spacing (mm)	0.1 [42]
ϵ	Emissivity	0.35
h	Heat transfer coefficient (w/m ² /K)	12.7

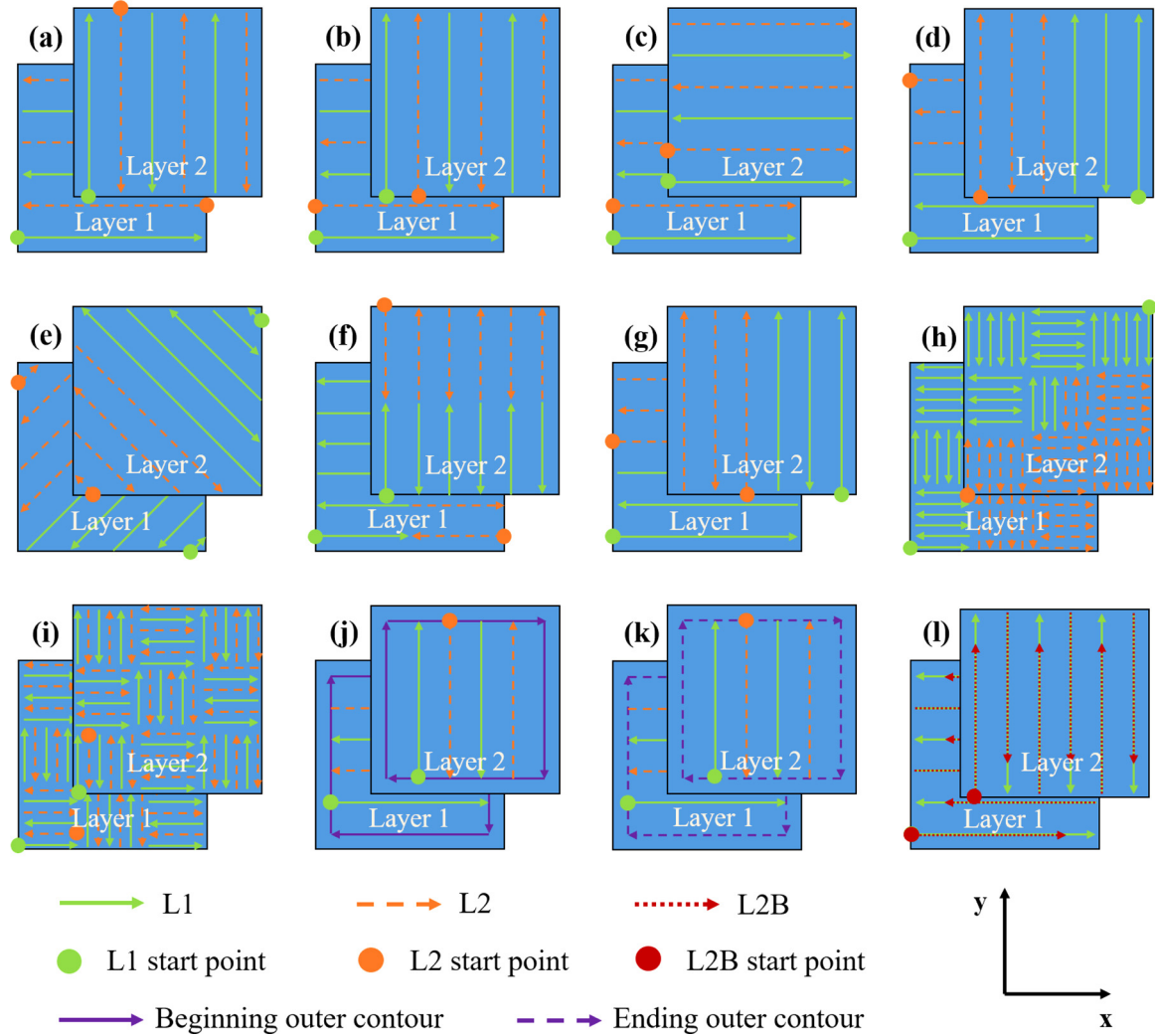


Fig. 2. Different scanning strategies arrangement for dual laser beams PBF-MLB: a) Opposite S scanning. b) Parallel S 90° rotation scanning. c) Parallel S no rotation scanning. d) 0° approaching beam scanning. e) 45° rotation approaching beam scanning. f) Opposite halves scanning. g) Parallel halves scanning. h) Island approaching beam scanning. i) Island mixed scanning. j) Beginning contour fill. k) Ending contour fill. l) Delayed laser scanning. For illustration, the hatch spacing has been magnified.

for the larger number of laser beams PBF-MLB, which was due to the lower temperature gradient [20] caused by the multiple laser beams.

Temperature gradients (calculated per [20]) were sampled at two locations- the last element of the melt pool in the first and the last element in the last layer, at the same time i.e. when the final layer is complete. At the top layer, the gradients were 48.18 K/μm, 56.47 K/μm, 59.28 K/μm and 65.863 K/μm, and at the bottom layer the gradients were 6.63 K/μm, 9.64 K/μm, 9.61 K/μm and 8.61 K/μm, for the single laser beam, dual laser beams, four laser beams and thirty laser beams models, respectively. The increase in temperature gradient causes a higher stress at the top surface for the larger number of laser beams. It is interesting to note that by the time the laser has completed the final layer, the trend in temperature gradients at the interface of the part and build plate had almost reversed resulting in a lower stress at the build plate interface for multi-laser systems.

Processes with additional laser beams apply more energy at a given time and thus resulting in more heat to be conducted to the previous solidified layers and increasing the overall temperature of the powder bed [11]. For multiple laser beams PBF-MLB, the part was fabricated faster and there was less time for the heat to be dispersed compared with the single laser beam PBF-LB, resulting in a higher average temperature, initially a higher temperature gradient in the most recent built layer, but overall a more uniform cooling and eventual lower

temperature gradient in older previous solidified layers.

The effectiveness of PBF-MLB over single beam PBF-LB in terms of shorter build times, faster building rates, lower residual stresses and deflections, which is detailed at Table 3. Though the laser exposure time for four laser beams PBF-MLB is 25 % of the single beam PBF-LB [8,43], the building rate is just 2.74 times faster due to time overhead required for material spreading for each layer. The peak temperature increases with the number of laser beams, as there is more heat input at any given time and less time for the energy to be lost through conduction, convection or radiation, therefore, the temperature gradient decreases and the cooling rate slows down, causing a reduction in residual stress and deflection.

3.2. Influence of scanning strategy on the thermal profile

The computational modelling results for the temperature fields of the additively manufactured parts using various scanning strategies are demonstrated in Fig. 4. The maximum temperature (called peak temperature in this paper) of the layer of the material in relation to respective scanning strategies is shown by using the value on top of each image in Fig. 4. Different scanning strategies obtained their peak temperatures at different time points, and the temperature contours at the half of the total time needed for a layer scanning of the top layer was

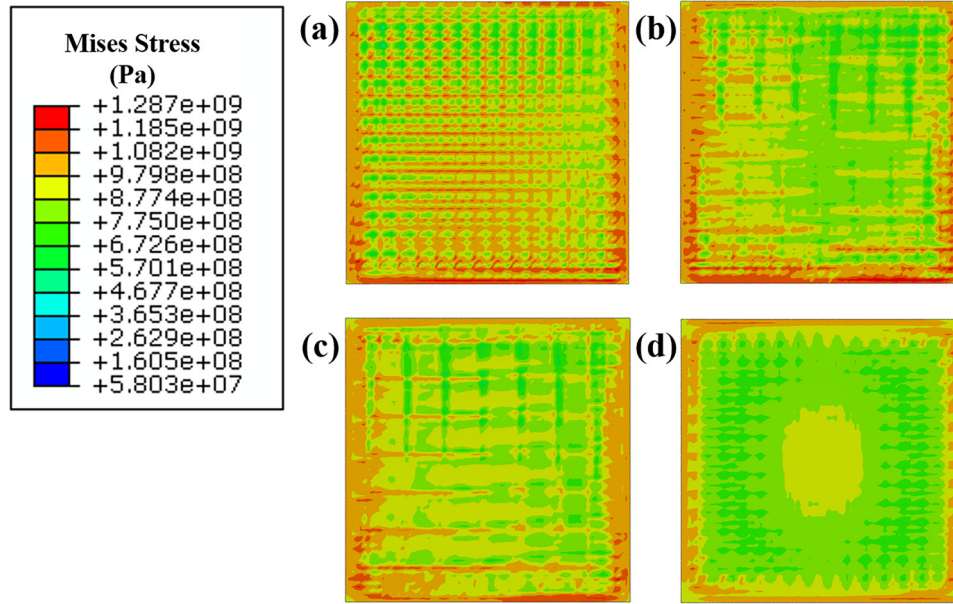


Fig. 3. Final Von Mises residual stress at the interface of the part and base plate (before releasing base plate constraints) for different numbers of laser beams: a) Single laser beam. b) Dual laser beams. c) Four laser beams. d) Thirty laser beams.

Table 3

Comparison of multiple laser beams PBF-MLB with the single beam PBF-LB in this study.

Items	PBF-LB (single laser)	PBF-MLB (dual lasers)	PBF-MLB (four lasers)	PBF-MLB (thirty lasers)
Laser exposure time	100 % (0.45 s)	~ 50 % (0.225 s)	~ 25 % (0.1125s)	~ 3.33 % (0.015 s)
Peak temperature	100 % (4145 K)	101.71 % (4216 K)	103.04 % (4271 K)	105.16 % (4359 K)
Residual stress	100 % (1438 MPa)	95.90 % (1379 MPa)	90.61 % (1303 MPa)	83.5 % (1201 MPa)
Deflection	100 % (3.61 μ m)	96.68 % (3.49 μ m)	90.30 % (3.26 μ m)	62.33 % (2.25 μ m)

plotted to show the process temperature fields of all the scanning strategies [16]. Comparing all the scanning strategies, the island mixed scanning strategy (Figs. 2i and 4 i) obtained the highest peak temperature, which is due to the dual laser beams locally scanning a very small area of material simultaneously. This conclusion agrees with the PBF-LB research outcomes by Cheng et al. [16]. While the delayed laser scanning (Figs. 2l and 4 l) obtained the second highest temperature which may due to the shorter distance (90 μ m) of the two beams compared with other scanning cases (100 μ m), even though the secondary laser beam was with a reduced laser power. The parallel halves

scanning strategy (Fig. 2g), in which the two laser beams were maintained farthest apart from each other, obtained the lowest peak temperature.

3.3. Influence of scanning strategy on the residual stress

3.3.1. Final residual stress of the part manufactured by using the opposite S scanning (see Fig. 2a)

The final residual stress refers to the residual stress of the part after the build manufacture is completed and has been allowed to cool for

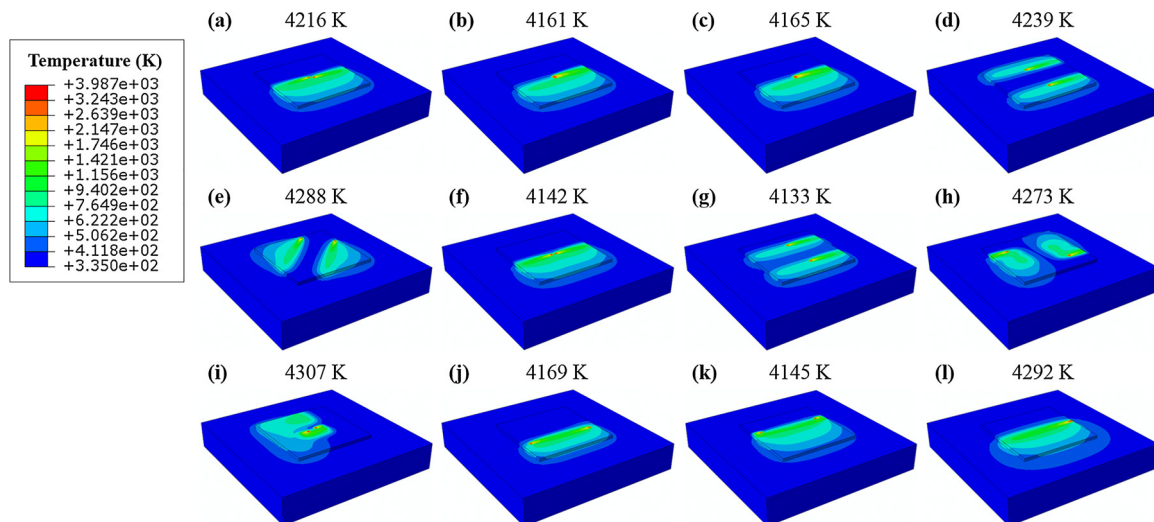


Fig. 4. Process temperature contours of the twelve scanning strategies (logarithmic scale) before releasing base plate constraints.

2000s. The multiple cut views of residual stress (Von Mises stress, x direction stress S11, y direction stress S22, z direction stress S33, maximum principal stress and minimum principal stress) are presented in Fig. 5. The x (transverse direction) and y (longitudinal) directions have been shown in Fig. 2 and z direction (build direction) is vertical to the x-y plane. Fig. 5a shows that the maximum Von Mises stress occurs at the interface between the part and the base plate, indicated by the red arrows (Fig. 5a). At the interface, the Von Mises stress is higher at the edges than that at the centre. Fig. 5b illustrates that the x direction stress centre is tensile stress while stress at the two edges in the x direction is compressive. For the y direction stress S22, the highest tensile stress occurs at the central areas of the interface (Fig. 5c). The high S11 and S22 residual stress located at the interface between the part and the base plate was also found in the computation for the single laser beam PBF-LB process [16]. Fig. 5d shows the highest S33 can be found at the interface between the part and the base plate. The maximum principal tensile stress also occurs at the interface (Fig. 5e). While the least minimum principal stress occurs at the top surface of the part and the minimum principal stress has the highest value along the trajectories of the laser beam scanning (Fig. 5f). The high residual stress at the interface between the part and the base plate has potential to cause cracking of the manufactured part either during manufacturing (hot cracking) or in service (cold cracking).

3.3.2. The maximum principal stress

Fig. 6 shows the modelling results of the maximum principal stress contours for the twelve different parts that were manufactured using respective scanning strategies before releasing constraints at the bottom of base plate (Fig. 2). The sub-figures in Fig. 6 and contour figures thereafter in this work shared the same legend scale in each individual figure. The scanning patterns (both the scan directions and the scan vectors) can determine the stress distributions (Fig. 6) and the average (Fig. 7) of the maximum principal stress. The results show the parallel S no rotation scanning (Figs. 2c, 6 c and 7 c) produced the highest average maximum principal stress (900.57 MPa). The parallel S 90° rotation scanning (Figs. 2b and 6 b) resulted in a lower and more uniform distribution of the maximum principal stress than that of the parallel S no rotation scanning (Figs. 2c and 6 c), demonstrating the importance of rotating the laser scan direction by 90° after finishing

each layer. The delayed laser scanning (Figs. 2l, 6 l and 7 l) produced the lowest average maximum principal stress (772.30 MPa) among all the scanning strategies and it can be seen that the secondary delayed laser helps to mitigate the residual stress. Fig. 7 illustrates 45° approaching beam scanning (Fig. 7e) caused a marginal 3.12 % lower residual stress than 0° approaching beam scanning (Fig. 7d). Island scanning (Fig. 6h and i) produced a higher residual stress at the island interface than the island central areas. It can be seen that regardless of the specific laser scanning strategy, high maximum principal stress always occurs at the interface between the part and the base plate. At the central area of the part, the residual maximum principal stress is always higher than that at the border of the part. It can be seen from Fig. 6 that the maximum principal stress of both beginning (Fig. 6j) and ending (Fig. 6k) contour fill scanning is slightly lower than that of opposite S scanning (Fig. 6a), which indicates that the contour fill scanning strategy is beneficial for mitigating the residual stress. The residual stress of opposite S scanning strategy (Fig. 6a) is higher than that of Parallel S 90° rotation scanning strategy (Fig. 6b) due to the peak temperature of the opposite S scanning strategy (Fig. 4a, 4216 K) being higher than that of the Parallel S 90° rotation scanning strategy (Fig. 4b, 4161 K). The case (a) and case (b) had larger residual stresses than the case (d), case (f) and case (g), which indicates that laser proximity influences the final residual stress that two laser beams close to each other obtained larger residual stress than two laser beams kept far away. This is assumed to be caused by laser beams close to each other had higher temperatures (Fig. 4) and thus had larger temperature gradients.

3.3.3. The direction residual stress

The S11 residual stress before releasing constraints at the bottom of base plate is presented in Fig. 8 by using the contour plots and the average S11 (of all the nodes at the top surface) is shown in Fig. 7. Fig. 8 shows the S11 stress at the central area of the part is always higher than that at the border of the part. The results from Figs. 7c and 8c show the maximum S11 stress (844.85 MPa) occurs at parallel S no rotation scanning, which is 1.28 % larger than the parallel S scanning with 90° layer rotation (Fig. 8b). Figs. 8e and 7 e illustrate the 45° rotation approaching beam scanning (Fig. 2e) resulted in the minimum average S11 residual stress (624.14 MPa) and the approaching beam

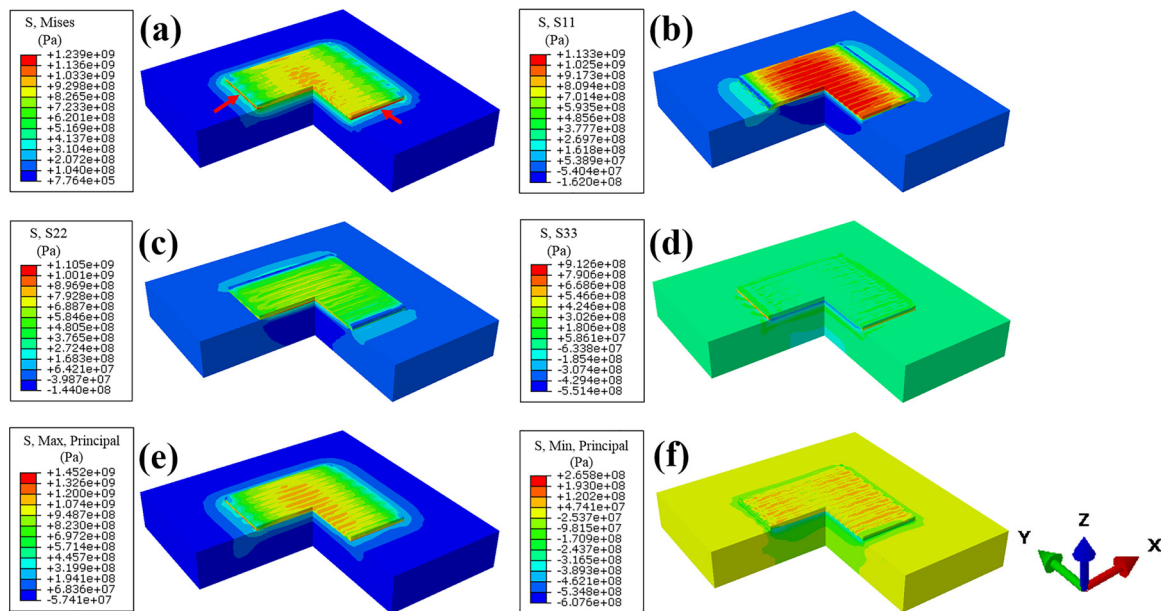


Fig. 5. Multiple cut views of scanning case (a) (see Fig. 2a) after cooling down for 2000s (before releasing base plate constraints): a) Von Mises stress. b) S11 residual stress along the x direction. c) S22 residual stress along the y direction. d) S33 residual stress along the z direction. e) The maximum principal stress. f) The minimum principal stress.

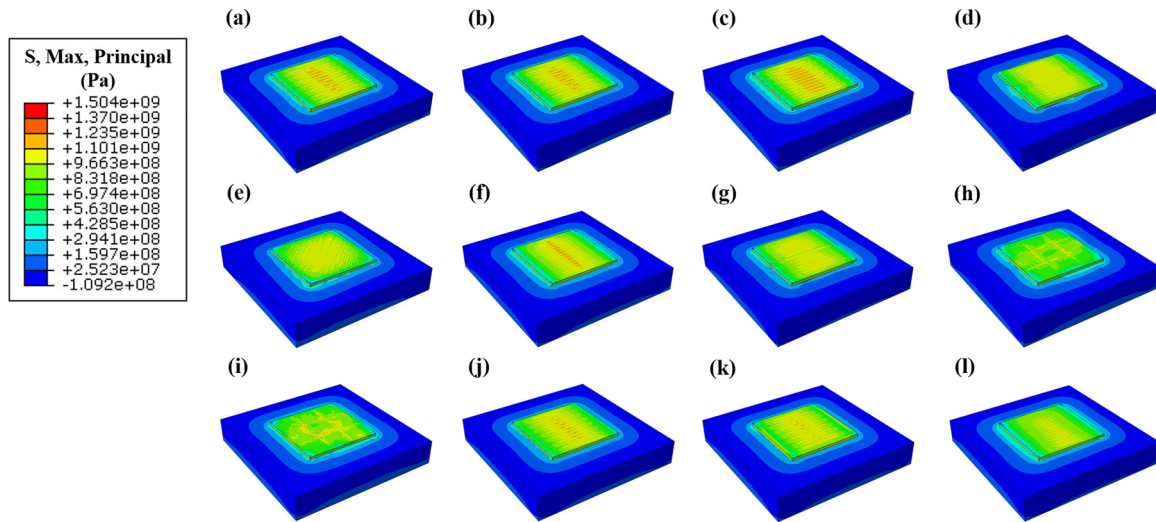


Fig. 6. The final maximum principal stress for the different scanning strategies before releasing base plate constraints.

scanning (Fig. 7h) generated the second lowest average S11 residual stress (671.23 MPa). The lowest average S11 of the 45° rotation approaching beam scanning (Fig. 8e) is 26.12 % lower than that of the highest S11 of the parallel *S* no rotation scanning (Fig. 8c). The S11 stress distribution pattern by scanning strategies of (Fig. 2a–d and f–g) are consistent with results of literature [21,44]. The delayed laser scanning (Fig. 8l) obtained significant reduced (8.35 % lower) S11 residual stress compared with the opposite *S* 90° rotation scanning (Fig. 8a).

Fig. 9 shows the *y* direction residual stress S22 for all the twelve scanning strategies. Being different with the results for S11 (Fig. 8), the S22 for the island scanning (Fig. 9i, Fig. 2i and Fig. 9h, Fig. 2h) and the 45° rotation approaching beam scanning (Figs. 9e and 2e) produced higher S22 residual stresses than other scanning strategies. The island mixed scanning (Figs. 9i and 10 i) had the maximum S22 stress of 710.81 MPa due to that the scan direction of the islands rotates 90° for subsequent island. The 45° rotation approaching beam scanning (Figs. 9e, 2e) had the minimum S11 stress but had comparative higher S22 stress may be caused by the comparatively more uniform temperature gradient variance in *x* and *y* directions [16]. The scanning cases (e), case (h) and case (i) had larger S22 stress at the part central fields than the borders. Except case (e), case (h) and case (i), all other

scanning strategies had similar low S22 stress distribution and quite uniform S22 stress distribution at the central areas and the borders.

Fig. 10 shows the maximum residual stress (S11, S22 and maximum principal stress) statistics for all the twelve scanning strategies. Combining the results of Figs. 7, 8 and 10, the parallel *S* no rotation scanning (Figs. 2c and 10 c) obtained the highest maximum principal stress (1504 MPa) and highest average maximum principal stress while the 45° rotation approaching beam scanning (Fig. 2e) had both the least S11 average distribution and the least maximum *x* direction residual stress (1005 MPa). The least maximum principal stress occurs at the delayed laser scanning of 1447 MPa (Fig. 10l), which validated the assumption that the addition of a second laser could reduce stress. Fig. 10 states the maximum S22 (1154 MPa) occurred on the island mixed scanning (case i) and the delayed laser scanning (Fig. 10l) had the lowest maximum S22 residual stress (861.5 MPa) among all scanning cases (Fig. 2). The highest S22 stress (Fig. 10i) was 33.95 % higher than the lowest maximum S22 stress (Fig. 10l). The ending contour fill (Fig. 10k) caused a 16 MPa lower maximum principal stress than the beginning contour fill (Fig. 10j), which indicates the ending contour fill scanning strategy could be beneficial for reducing residual stress.

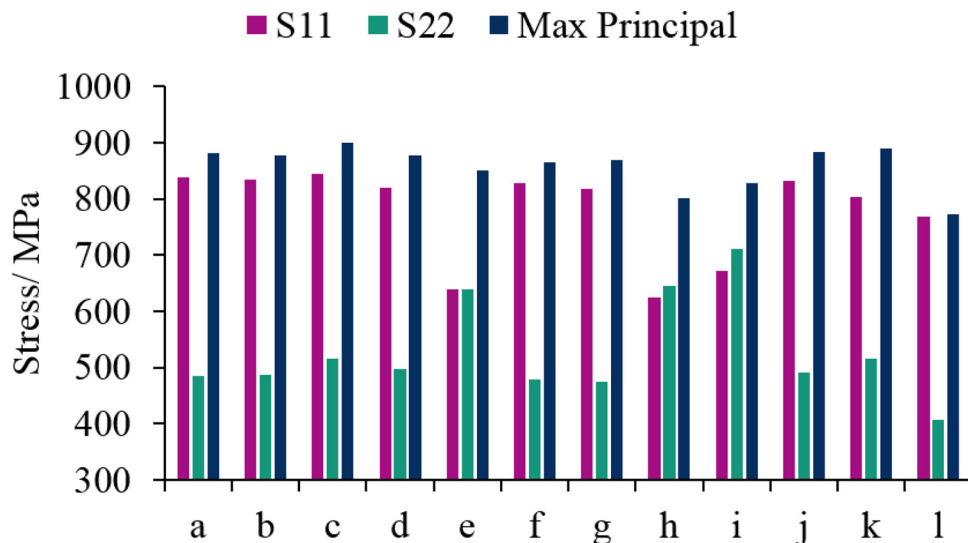


Fig. 7. Final average residual stress statistics for the different scanning strategies before releasing base plate constraints (S11, S22 and maximum principal stress).

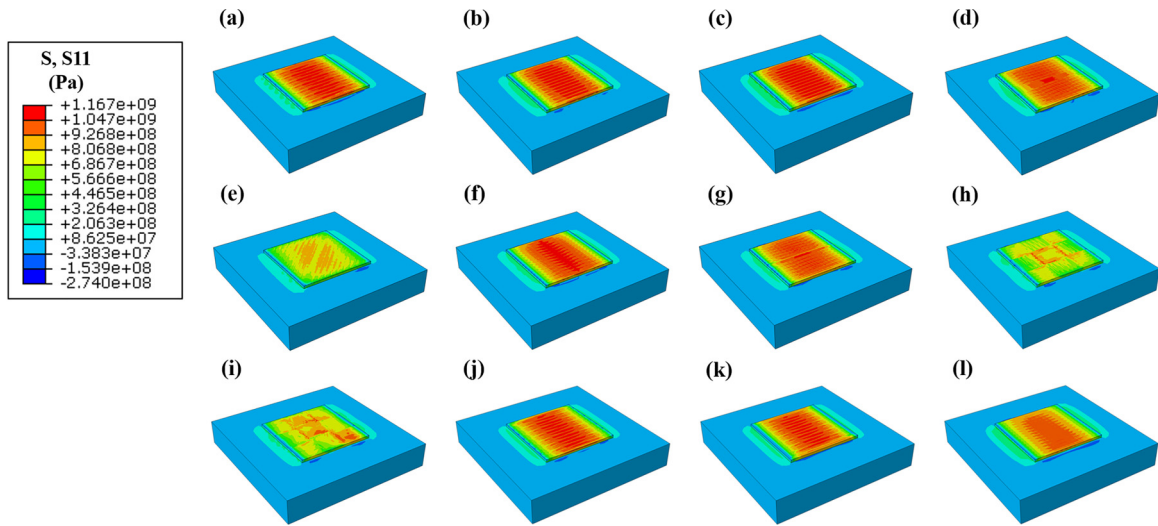


Fig. 8. Final residual stress along the x direction S11 for the different scanning strategies before releasing base plate constraints.

3.4. Influence of scanning strategy on the Z direction deflection

3.4.1. Z direction deflection for the component

Fig. 11 illustrates the final z direction deflection of the parts for the six typical scanning strategies (Fig. 2b–e, i and l). The deflection contours show significant influences of the scanning strategies on the final z direction deflection. The z direction deflection distributions resemble the laser beams scanning paths (Fig. 11i). It can be seen that the negative z direction deflection is predicted in the laser beam path, with positive z direction deflection predicted between neighbouring laser paths (Fig. 11). A similar deformation was observed in the literature [19,45,46]. The z direction deflection of parallel S 90° rotation scanning (Fig. 11b, Fig. 2b) is a lower than the parallel S no rotation scanning (Figs. 11c and 2 c), which is consistent with the maximum principal stress results (Fig. 6).

Fig. 12 shows the maximum z direction deflection of the parts that were manufactured using different scanning strategies. It can be found that the largest maximum z direction deflection occurred at the island mixed scanning (5.47 μm) (Fig. 12i) was 152 % higher than the lowest maximum z direction deflection that was occurred at the 45° rotation approaching beam scanning (2.17 μm) (Fig. 12e). The island mixed scanning had the maximum z direction deflection, which was due to the energy concentration of the two laser beams focused on one island

simultaneously and the residual heat effect [16]. The peak temperature of the island mixed scanning (Fig. 4i) is the highest among all cases which also causes a higher temperature gradient that further increased the deflection. The case (j) and case (k) had higher z direction deflections than case (a), which indicates that the contour fill scanning leads to higher deflection than no contour fill. A possible reason is that case (j) and case (k) had shorter scanning vectors than case (a), which caused more energy concentrated at the part central areas during the same time period. It can be seen from Fig. 12 that the case (a) and case (b) had larger maximum z direction deflections than case (d), case (f) and case (g), which indicates that laser proximity also influenced the z direction deflection that laser beams closer to each other got larger z direction deflection, which is the consistent with the influences of laser proximity to the temperatures and residual stresses (Fig. 6). There were minor differences of the maximum z deflections for the 0° approaching beam scanning (case (d)), opposite halves scanning (case (f)), parallel halves scanning (case (g)) due to that the dual laser beams kept far away from each other during scanning.

3.4.2. Z direction deflection and distribution of the base plate

Fig. 13a illustrates the maximum z direction deflection of the base plate for all the scanning strategies and Fig. 13b shows the contour plot for z direction deflection of the base plate for scanning strategy case a.

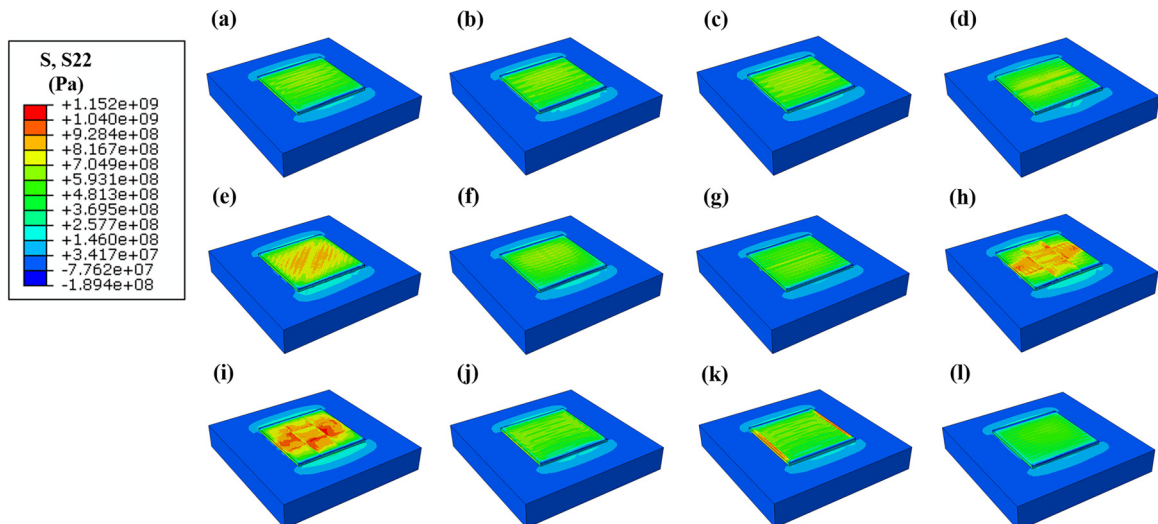


Fig. 9. Final residual stress along the y direction S22 for the different scanning strategies before releasing base plate constraints.

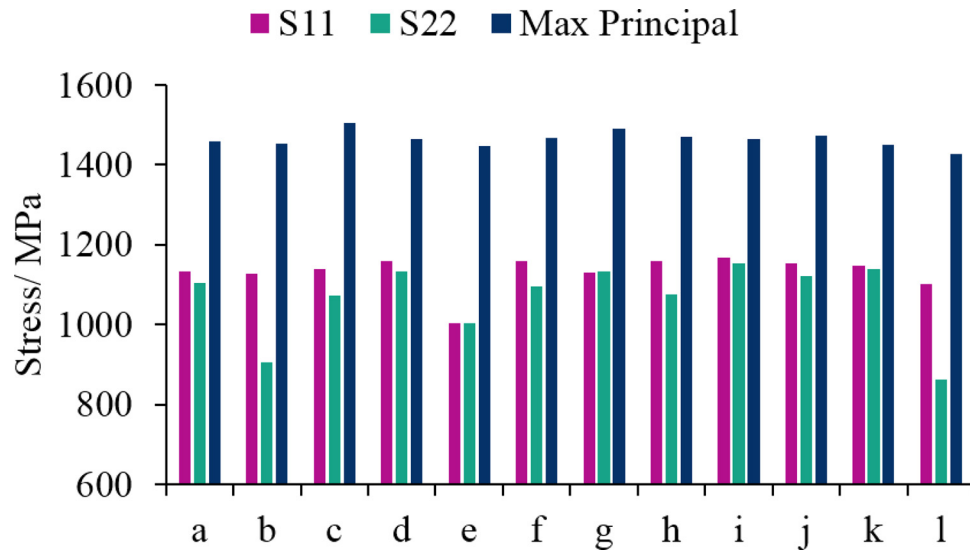


Fig. 10. Final maximum residual stress (S11, S22 and maximum principal stress) statistics for the different scanning strategies before releasing base plate constraints.

In Fig. 13a, it can be seen that the 0° approaching beam scanning (case d), the 45° rotation approaching beam scanning (case e), the island approaching beam scanning (case h) and the island mixed scanning (case i) resulted in comparatively higher maximum z direction deflection on the base plate than other cases. The difference of z direction deflection between the part and the base plate illustrates the final component's deflection is not only determined by the base plate deflection, but also influenced by the part deposition process. Fig. 13b states the base plate deformation focused on the edges of the base plate interface contacted with the part as well as the part central area on the base plate. The laser scanning paths centres also deformed more than other fields. A similar base plate z direction deflection was found in [47].

3.5. Influence of base plate constraints on the residual stress and deflection

3.5.1. Residual stress for the part

The final maximum residual stress (S11, S22 and principal stress)

statistics for the different scanning strategies after releasing constraints at the bottom of the base plate are shown in Fig. 14. Releasing these constraints resulted in some reordering of the scanning strategies in terms of stress and deflection behaviour. The lowest S11 residual stress occurred at the 45° rotation approaching beam scanning (Fig. 14e) and the delayed laser scanning (Fig. 14l) had the lowest S22 residual stress, which is consistent with the modelling results before releasing the base plate constraints. However, the largest maximum principal stress occurred at the parallel halves scanning (Fig. 10g) after releasing the base plate constraints while the largest maximum principal stress occurred at case (c) before releasing constraints of the base plate. Directly comparable results were obtained for the maximum principal stress before (Fig. 10) and after (Fig. 14) releasing constraints at the bottom of the base plate for this thin plate structure. In this study, the part removal process (e.g. wire electrical discharge machining) and post heat treatments were not included.

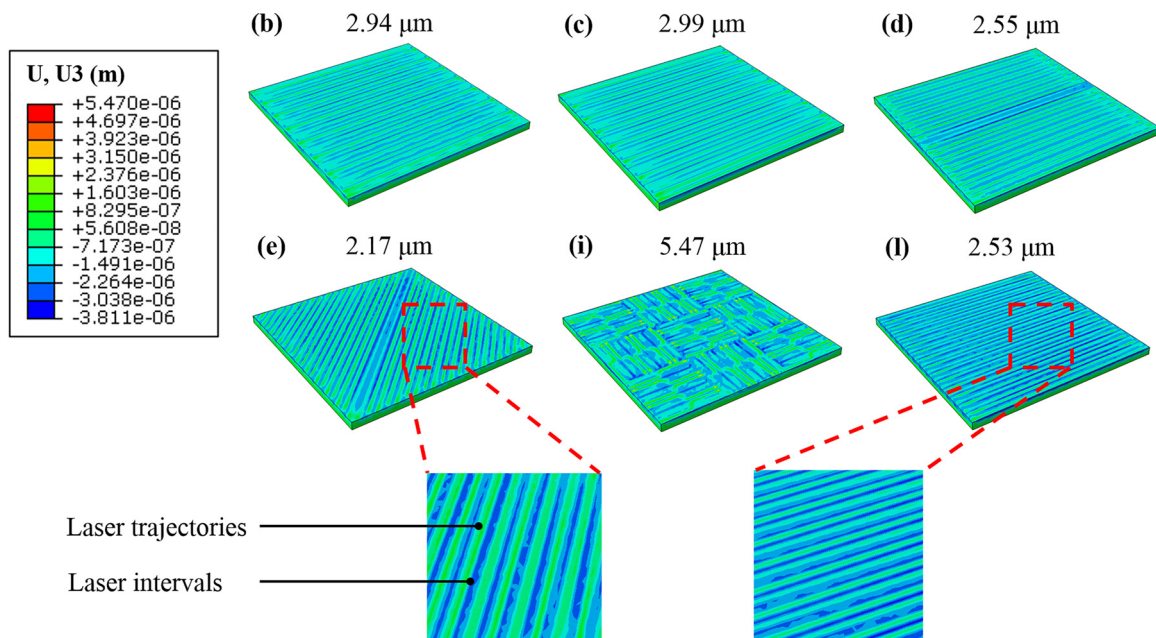


Fig. 11. Final z direction deflections for the different scanning strategies before releasing base plate constraints.

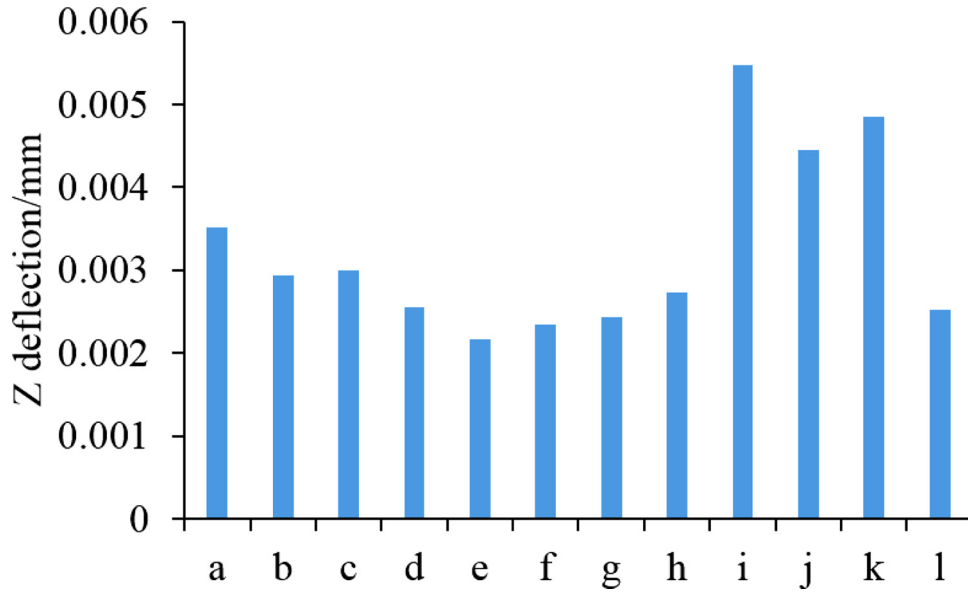


Fig. 12. Maximum z direction deflection statistics for all cases before releasing base plate constraints.

3.5.2. Z direction deflection for the part

The z direction deflections of the part for the different scanning strategies after releasing constraints at the bottom of the base plate are shown in Fig. 15. The z direction deflections became negative after releasing constraints at the bottom of the base plate (Fig. 11). This is caused by the negative deflections at the base plate [21,38] and there were less constraints at the bottom surface of the base plate and thus the part has more freedoms to deform. The Parallel halves scanning (case g) had significant deflection compared with other scanning cases. The 0° approaching beam scanning (case d) had the least z direction deflection. The parallel S no rotation scanning (case c) had larger z direction deflection compared with parallel S 90° rotation scanning (case b), which is consistent with the experimental measurements that single x direction scanning caused larger deformation compared with the 90° rotation scanning strategy [48].

4. Conclusions

Fully coupled thermo-mechanical modelling was employed to computationally predict the thermal process and the residual stress of Ti6Al4V alloy parts that were manufactured by dual laser beams PBF-MLB. Twelve different scanning strategies were used in the modelling. The results show scanning strategies influence the final part residual stress and deformation significantly for the PBF-MLB. Based on the above thermal, residual stress and deflection results, the general guidelines for optimizing the scanning strategies on dual laser beams PBF-MLB are as follows:

- The lower residual stress and z direction deflection were formed for the larger number of laser beams PBF-MLB.
- To obtain the lowest peak temperature, the two laser beams need to be spaced apart from each other. Laser proximity influences peak temperatures, which is due to that the energy concentration and shorter scanning vectors led to higher peak temperatures.
- The parallel S no rotation scanning (Fig. 2c) produced the highest maximum residual stress and larger stress than the parallel S 90° rotation scanning (Fig. 2b), which indicates that 90° layer rotation after each layer is also necessary for minimising the final residual stress in PBF-MLB. This is likely due to perpendicular laser trajectories between each layers balancing the directional residual stresses.
- The delayed scanning strategy (Fig. 2l) is beneficial for reducing residual stress because the secondary delayed laser slows down the cooling speed and thus mitigate the residual stress.
- The 45° rotation approaching beam scanning obtained the second least residual stress and the parallel S no rotation scanning should be avoided as it resulted the highest residual stress.
- The island mixed scanning strategy gives the highest peak temperature and the parallel halves scanning strategy obtained the lowest peak temperature.
- The 45° rotation approaching beam scanning had the lowest z direction deflection.
- The results of higher residual stresses occurred along the interface between the base plate and the part for the dual laser beams PBF-MLB simulation is consistent with the single laser beam PBF-LB

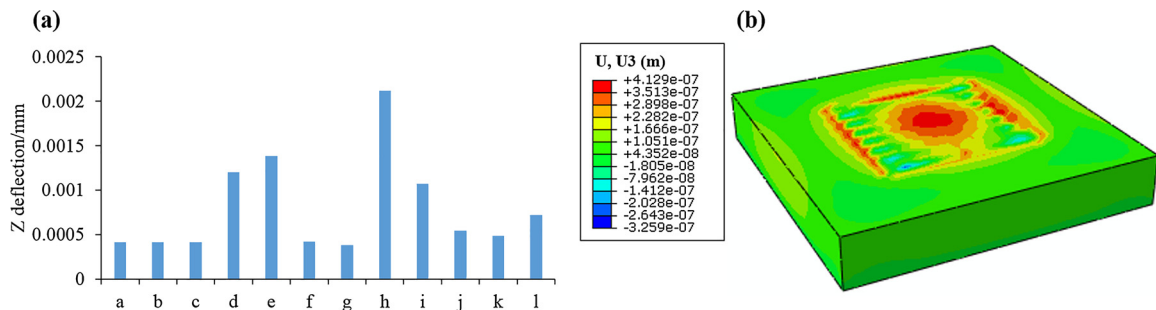


Fig. 13. Z direction deflection on the base plate before releasing base plate constraints: a) Maximum z direction deflection statistics for the twelve scanning strategies. b) Contour of the opposite S scanning pattern (a).

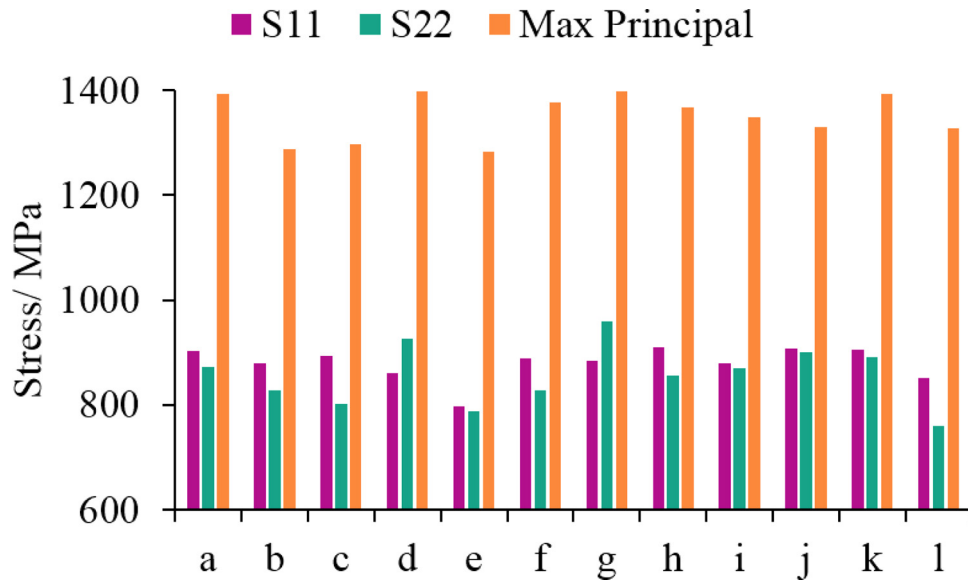


Fig. 14. Final maximum residual stress (S11, S22 and maximum principal stress) statistics for the different scanning strategies after releasing constraints at the bottom of base plate.

simulation.

- After releasing constraints at the bottom of the base plate, a decrease in residual stress was found. The parallel halves scanning had the largest residual stress (maximum principal stress) and the lowest stress was occurred at the 45° rotation approaching beam scanning.
- After releasing constraints of the base plate, the 0° approaching beam scanning had the least z direction deflection.

Future work of this research will require deflection and residual stress measurements for model validation. The comparison of the 45° rotation approaching beam scanning by the dual laser beams PBF-MLB with the single laser beam PBF-LB modelling (e.g. 45° inclining scanning) is also needed. Furthermore, four laser beams PBF-MLB simulation scanning strategies are necessary to be investigated by both modelling and experiment. Finally, the effect of reaching different peak temperatures and different cooling rates on the microstructural evolution (grain growth and phase) should also be considered in process optimisation studies [49].

Declaration of Competing Interest

None.

Acknowledgments

The authors grateful acknowledge the funding by the Irish Research Council (GOIPG/2018/2488) under the Government of Ireland Postgraduate Programme, the Hardiman Research Scholarship and College of Engineering and Informatics of NUI Galway. The work is also supported in part by a research grant from Science Foundation Ireland (SFI) under Grant Number 16/RC/3872 and is co-funded under the European Regional Development Fund. The authors wish to acknowledge the Irish Centre for High-End Computing (ICHEC) for the provision of computational facilities and support.

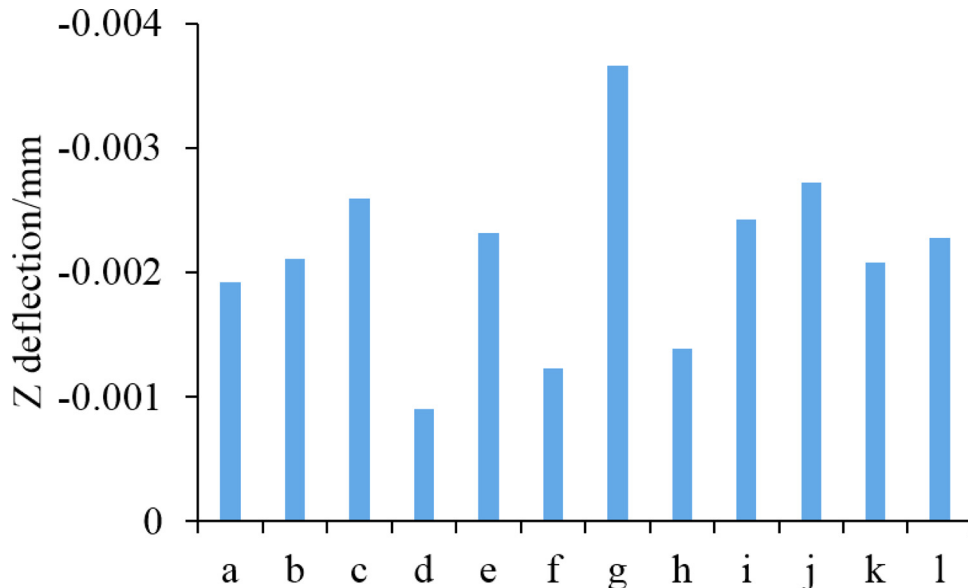


Fig. 15. The final z direction deflection statistics for the different scanning strategies after releasing base plate constraints.

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